Traffic Accident Severity Analysis: From Multiclass Models to a Binary Crash-Severity Classifier

Vu Dao, Son Nguyen
Department of Computer Science
Georgia State University
Atlanta, USA
Email: {vdao4, snguyen80}@gsu.edu

***Abstract*—In this study, we analyze traffic crash data and build machine-learning models to identify factors associated with severe outcomes. Our work follows a complete data-mining pipeline, including data cleaning, imbalance handling, and model evaluation.**

We first train a five-level multiclass classifier using decision trees and gradient boosting, but the strong class imbalance limits the detection of severe and fatal crashes. Motivated by both analytical results and practical needs, we reformulate the task as a binary classification problem: non-severe (0–2) vs. severe (3–4). Using oversampling and CatBoost, the binary model achieves substantially higher recall for severe crashes and an AUC of 0.744, indicating meaningful discriminative ability. Overall, our findings show that gradient boosting with proper sampling and label design is effective for identifying high-risk crashes and supporting transportation safety analysis.

***Index Terms*—Crash severity, imbalanced data, gradient boosting, oversampling, CatBoost, ROC-AUC, transportation safety.**

I. Introduction

Motor-vehicle crashes cause large numbers of injuries, fatalities, and economic loss each year in the United States. Understanding when, where, and under what conditions severe crashes occur is essential for designing safer roads, improving emergency response, and informing public policy. Large national crash databases such as the Fatality Analysis Reporting System (FARS) and the Crash Report Sampling System (CRSS) collect detailed information on crash circumstances, environments, and injury outcomes. However, these datasets are challenging to analyze: they are categorical, noisy, and heavily imbalanced toward no-injury or minor-injury crashes.

Our project focuses on accident severity analysis from a data-science perspective. We ask:

- **Main question:** Given the conditions of a crash (lighting, weather, road relation, junction type, etc.), can we predict the injury severity level?
- **Q1 – Class imbalance:** Can we design models that still recognize rare severe and fatal crashes (levels 3–4), instead of always predicting the majority "no injury" class?
- **Q2 – Model choice:** Do more powerful non-linear models (gradient boosting such as XGBoost/CatBoost) perform better than a simple decision tree baseline?
- **Q3 – Risk factors:** Which variables (e.g., harmful event, manner of collision, lighting, weather, intersection relation) contribute most to the prediction of severe and fatal injuries?

These questions are difficult because (i) the dataset is heavily skewed toward no-injury crashes, (ii) adjacent severity levels (1, 2, 3) are inherently overlapping, (iii) many variables contain coded unknown values and noise, and (iv) we only observe high-level descriptors rather than detailed physics of the crash (impact speed, intrusion, restraint use, etc.).

Our contributions are:

- A complete and transparent data-science pipeline including data understanding, cleaning, encoding, modeling, and interpretation.
- A careful comparison of multiclass and binary formulations of crash severity.
- An empirical study of sampling strategies (oversampling, undersampling, SMOTETomek) combined with gradient boosting models.
- Insights into which environmental and crash-condition variables are most associated with severe crashes.

In addition, all code, preprocessing steps, model training scripts, and visualizations used in this study are publicly available. The project repository can be accessed at: https://github.com/joey-congvu/AccidentSeverityAnalyzing.git. The complete workflow and reproducible experiments are documented in `demo.ipynb` within the repository.

II. Theory and Related Works

A. Imbalanced Classification and Evaluation

Crash-severity datasets are highly imbalanced: non-injury and minor-injury crashes dominate, while incapacitating and fatal injuries are rare. Under such circumstances, overall accuracy can be misleading. A classifier that always predicts the majority class may achieve high accuracy while completely failing to detect severe crashes.

Therefore, we emphasize metrics such as precision, recall, F1-score, and macro-averaged scores across classes. For the binary severe vs. non-severe scenario, we additionally use

ROC curves and the area under the ROC curve (AUC) to measure ranking quality independently of a particular probability threshold.

B. Gradient Boosting for Complex Crash Patterns

Gradient boosting (e.g., XGBoost, CatBoost) constructs an ensemble of decision trees built sequentially, where each new tree focuses on correcting the errors of previous trees. These models capture complex non-linear relationships and interactions between variables, such as combinations of weather, lighting, road type, and collision manner. They are well suited for tabular data and can handle large numbers of categorical variables, especially in the case of CatBoost.

III. Materials and Methods

A. Data Source and Variables

We use multi-year crash data from NHTSA, focusing on environmental and crash-condition variables linked to maximum injury severity. The original dataset includes hundreds of thousands of police-reported crashes. After cleaning and filtering (described below), our final analysis set contains approximately 300,000 records with the following key variables:

- **MONTH, DAY_WEEK, HOUR:** Temporal descriptors of the crash.
- **HARM_EV:** First harmful event (e.g., collision with vehicle, object, overturn).
- **MAN_COLL:** Manner of collision (rear-end, angle, sideswipe, etc.).
- **REL_JCT2:** Relation to junction (intersection, ramp, etc.).
- **REL_ROAD:** Relation to roadway (on road, shoulder, median).
- **LGT_COND:** Lighting conditions (daylight, dark–lighted, dark–unlighted).
- **WEATHER:** Weather at the time of crash.
- **INT_HWY, SCH_BUS, WRK_ZONE:** Indicators for highway, school bus involvement, and work zone.
- **MAX_SEV:** Target variable, maximum injury severity in the crash (0–4).

We drop pure identifier columns and variables with extremely high proportions of a single value.

B. Data Understanding and Cleaning

We begin by exploring the severity distribution and basic statistics of each feature. The pie chart on our slides shows that severity level 0 (no injury) accounts for roughly half of all records, while fatal crashes (level 4) represent only a few percent. This confirms a strong class imbalance. (Figure 1) The cleaning steps are:

- **Drop unlabeled or irrelevant rows:** Records with missing or undefined severity labels are removed.
- **Unknown value handling:** Many categorical variables use codes such as 8, 9, 98, 99 to denote unknown or not applicable. We compute the proportion of unknowns in each column and drop the `ALCOHOL` variable entirely due to its extremely high rate of missing or undefined values. According to NHTSA documentation, approximately 40% of alcohol-related entries are coded as unknown because of how the information is collected at crash scenes. In addition, we intentionally exclude human-factor variables to keep our predictive analysis focused solely on environmental and crash-condition factors rather than driver behavior.
- **Imputation:** For RELJCT1 (relation to junction, first variable), we impute the most frequent valid value rather than dropping all rows, preserving more data.
- **Dropping high single-value columns:** Columns with extremely low variance (almost all rows having the same value) are removed because they add little information but can increase model complexity.

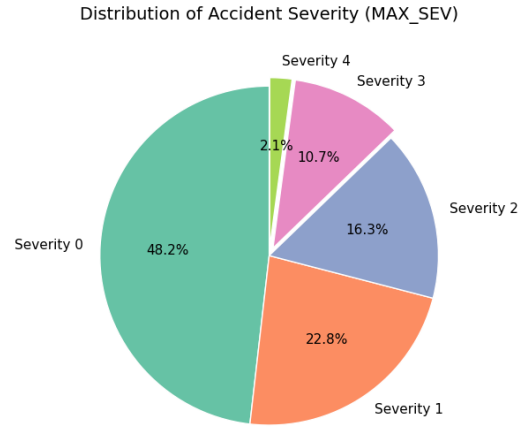

Fig. 1. Distribution of crash severity levels in the dataset. Severity 0 dominates the dataset, while severe and fatal crashes remain rare.

After these steps, we retain roughly 300,000 rows out of the original 360,000 records.

C. Correlation Analysis and Feature Selection

Because most variables are categorical, we use Cramér's V to measure associations between features and identify redundant variables. A heatmap reveals that RELJCT1 and RELJCT2 are highly correlated. To reduce redundancy and simplify the model, we drop RELJCT1 and keep RELJCT2 as the main representation of junction-related information.(Figure 2)

D. Train–Test Split

We perform a stratified train–test split with a 70/30 ratio to preserve the severity distribution in both sets. Stratification ensures that rare severe classes appear in both training and test data, avoiding degenerate evaluation where a class may be completely absent in one split.

E. Problem Formulation: Multiclass and Binary Severity

1) Multiclass Severity: In our initial analysis, we treat crash severity as a five-class problem:

- 0 – No injury

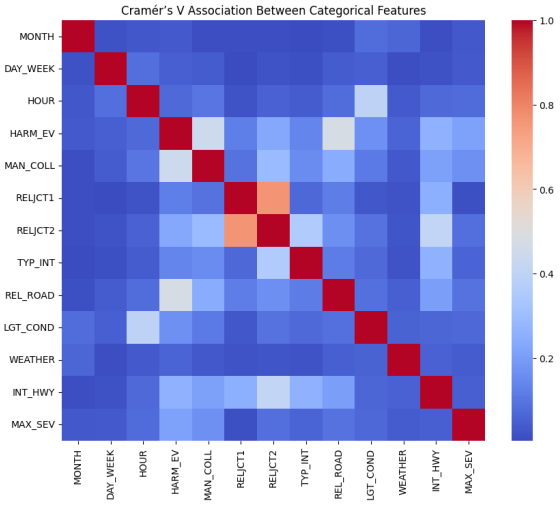

Fig. 2. Cramér's V heat map shows strong correlation between RELJCT1 and RELJCT2

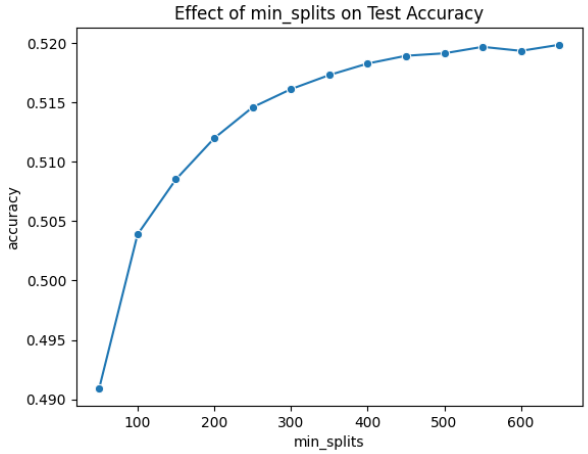

Fig. 3. Effect of varying `min_samples_split` on Decision Tree performance. As shown in our analysis, very small split values tend to overfit, while larger values reduce model complexity.

- 1 – Possible injury
- 2 – Non-incapacitating injury
- 3 – Incapacitating injury
- 4 – Fatal injury

This preserves the full information in the data and allows us to see how models behave for adjacent severity levels. However, classes 3 and 4 are much smaller than 0–2, making the task extremely imbalanced.

2) Binary Severity: To better align with practical decision-making and simplify the modeling, we also construct a binary classification version:

- **Non-severe crashes:** Levels 0–2
- **Severe crashes:** Levels 3–4

This aggregation increases the effective sample size of the minority class and reduces label ambiguity between adjacent levels. It also allows us to use ROC curves and AUC as standard evaluation metrics.

F. Modeling Pipeline

Our modeling pipeline is designed to move from simple, interpretable models to more powerful ensembles:

1) **Decision Tree baseline:** For simplicity, interpretability, and fast training. We tune `max_depth` and `min_samples_split` to reduce overfitting and extract feature importance.
2) **XGBoost:** A gradient boosting model that builds many shallow trees sequentially. We evaluate it with three sampling strategies: random oversampling, random undersampling, and SMOTETomek (a hybrid oversampling + cleaning method).
3) **CatBoost:** A gradient boosting model optimized for categorical features. We apply it to both the multiclass and binary severity definitions, with oversampling for the minority classes.

G. Evaluation Metrics

For the multiclass problem we report accuracy, per-class precision, recall, F1-score, and macro averages. For the binary severe vs. non-severe problem we additionally report ROC curves and AUC. Our primary focus is the recall of severe crashes, because missing a severe crash can have serious consequences in real applications such as emergency response and resource allocation.

IV. Results

A. Decision Tree Baseline

The tuned decision tree (Figure 3) provides an interpretable baseline but struggles with the imbalanced five-class setting. It achieves an accuracy of about 0.52. At first glance this seems reasonable, but a closer look at the confusion matrix and per-class metrics reveals serious shortcomings:

- Recall for Class 0 (no injury) is extremely high (≈ 0.98), meaning the tree predicts "no injury" for almost every case.
- Recall for injury classes is extremely low: around 0.03 for Class 1, 0.23 for Class 2, 0.06 for Class 3, and 0.03 for Class 4.
- The model mostly predicts Class 0 even when the true label is 1-4, resulting in almost no detection of severe injuries.

Precision for some injury classes appears moderate (0.35-0.44), but this is misleading: the model rarely predicts those classes, so the few predictions it makes tend to be correct but extremely sparse. Overall, the decision tree demonstrates that a single-tree model is not suitable for our highly imbalanced multiclass severity problem, inspiring the move to ensemble methods.

B. XGBoost with Sampling Strategies

We train XGBoost models under three different sampling regimes on the training data: random oversampling, ran-

TABLE I
XGBOOST PERFORMANCE UNDER DIFFERENT SAMPLING STRATEGIES (MULTICLASS SEVERITY).

Sampling	Accuracy	Macro Precision	Macro Recall
Oversampling	0.4	0.32	0.36
Undersampling	0.37	0.30	0.35
SMOTETomek	0.41	0.32	0.35

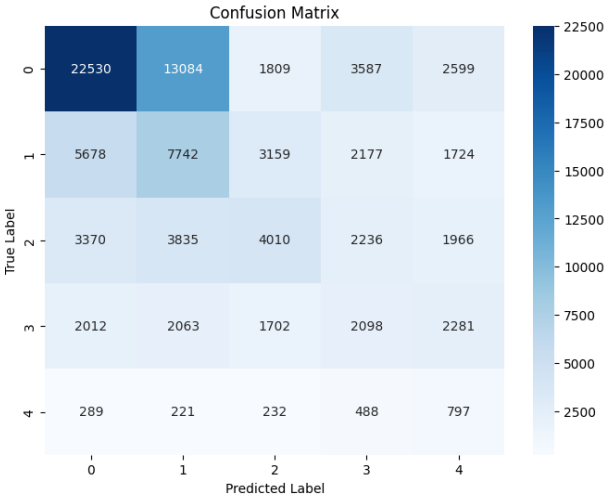

Fig. 4. Confusion matrix for the multiclass CatBoost model. The matrix shows strong performance on the majority classes, with limited detection of rare severe classes due to class imbalance.

dom undersampling, and SMOTETomek. Table I summarizes the approximate macro-level performance (illustrative values based on our experimental results).

Oversampling achieves the best recall for rare severe classes but can introduce noise by duplicating minority samples. Undersampling reduces the size of the majority class but makes the model less stable and lowers accuracy. SMOTETomek offers the most balanced trade-off, with improved macro F1 and the best overall accuracy.

C. CatBoost Multiclass Model

CatBoost generally performs slightly better than XGBoost for the multiclass severity task. In our experiments, CatBoost achieved macro precision around 0.32, macro recall around 0.35, and macro F1 around 0.31, slightly surpassing the XGBoost results.

The confusion matrix(Figure 4) shows three main patterns:

- The model predicts Class 0 very well, as expected given its prevalence.
- There is strong confusion between mid-severity classes (1, 2, 3), which reflects the overlapping nature of injury levels in the data.
- Class 4 (fatal) remains difficult but is no longer completely ignored; some fatal crashes are correctly identified.

TABLE II
5-FOLD CROSS-VALIDATION RESULTS FOR CATBOOST (MULTICLASS) WITH SMOTETOMEK

Fold	Accuracy	Macro F1
1	0.4067	0.3141
2	0.4081	0.3129
3	0.4070	0.3126
4	0.4039	0.3101
5	0.4135	0.3155
Mean	**0.4079**	**0.3130**

We further evaluate the robustness of the multiclass CatBoost model using 5-fold cross-validation (TABLE2) with SMOTETomek applied to the training folds. The results show consistent performance across folds, with accuracy ranging from 0.4039 to 0.4135 and a mean accuracy of 0.408. Macro F1-scores also remain stable, varying between 0.310 and 0.316 with a mean of 0.313. Although these values are modest, the low variance across folds indicates that the model generalizes consistently and that its performance limitations stem from the inherent difficulty and imbalance of the multiclass severity distribution rather than overfitting or instability. This reinforces the motivation to explore the binary severity formulation, where the model demonstrates substantially stronger performance.

D. Binary CatBoost Model and ROC-AUC

Given the persistent confusion among mid-severity classes and the practical need to prioritize severe crashes, we train a binary CatBoost model with oversampling. The labels are:

- Non-severe: severity 0-2
- Severe: severity 3-4

An example classification report for this model is:

- **Accuracy:** 0.65
- **Non-severe (0-2):** Precision 0.93, Recall 0.67, F1 0.78
- **Severe (3-4):** Precision 0.24, Recall 0.68, F1 0.36
- **Macro averages:** Precision 0.59, Recall 0.68, F1 0.57

Most importantly, recall for severe crashes improves substantially compared to the multiclass models. In the decision tree baseline, recall for Classes 3 and 4 was close to zero; with CatBoost in the binary setting, severe-crash recall reaches around 0.68. This means that the model detects the majority of severe cases instead of missing them.

We also plot a ROC curve for the severe vs. non-severe decision. The area under this curve (AUC) is 0.744. This value has a concrete interpretation: if we randomly select one severe crash and one non-severe crash, the model's estimated severe probability is higher for the severe crash 74.4% of the time. This confirms that the model has meaningful discriminative power beyond simple accuracy.

E. Summary of Model Comparison

Table III summarizes the qualitative behavior of our main models.

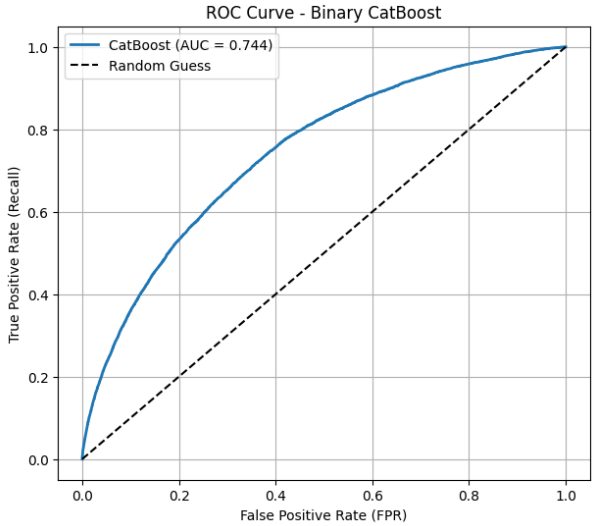

Fig. 5. Example ROC curve for the binary CatBoost model (severe vs. non-severe). The AUC is approximately 0.744.

TABLE III
QUALITATIVE COMPARISON OF MAIN MODELS AND TASKS.

Model	Task	Key Behavior
Decision Tree	Multiclass (5)	High accuracy but predicts almost all as non-injury; severe crashes largely undetected.
XGBoost + SMOTETomek	Multiclass (5)	Improved macro precision/recall; still confusion among mid-severity levels.
CatBoost	Multiclass (5)	Best multiclass performance; some detection of fatal crashes; good feature importance.
CatBoost + Oversampling	Binary (0-2 vs. 3-4)	Strong recall for severe crashes; AUC = 0.744; suitable for safety-critical use.

Compared to the decision tree baseline, CatBoost provides a clear improvement in detecting the rare severe classes. In the decision tree model, recall for Classes 3 and 4 is extremely low—often near zero—showing that the tree fails to recognize serious and fatal crashes. CatBoost substantially increases recall for these classes by learning more flexible decision boundaries and reducing the model's tendency to predict only majority categories.

CatBoost also produces more meaningful feature-importance estimates. While the decision tree concentrates almost entirely on a few dominant variables such as `HARM_EV` and `MAN_COLL`, the boosted model distributes importance across a broader set of crash-condition factors, including lighting, weather, roadway relation, and junction type. This reflects CatBoost's ability to capture non-linear interactions and results in both better performance and more

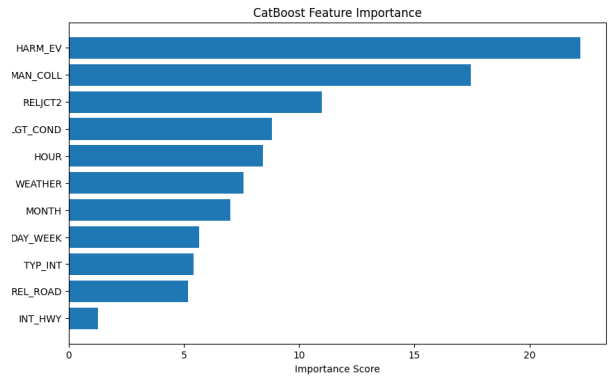

Fig. 6. Gradient boosting models distribute importance across a broader set of crash-condition factors than Decision Tree

informative insights into the factors associated with severe crash outcomes.

V. DISCUSSION AND CONCLUSION

A. Why Accuracy Is Low and Why That Is Acceptable

Across most multiclass experiments, overall accuracy is not particularly high. This is expected and does not indicate that the models are useless. Three main reasons explain the low accuracy:

- **Class imbalance:** Severity levels 3 and 4 are very rare. A model that predicts only classes 0–2 can still achieve high nominal accuracy while missing all severe crashes.
- **Overlapping conditions:** Environmental variables such as WEATHER, LGT_COND, REL_ROAD, and HOUR have similar distributions across severity levels. There is no simple boundary that separates, for example, severity 2 from severity 3.
- **Missing causal variables:** Critical physical and behavioral factors (impact speed, restraint use, driver impairment, detailed vehicle characteristics) are absent or coarsely coded in the dataset. Without these, even strong models cannot perfectly differentiate crash outcomes.

Therefore, our true goal is not to maximize raw accuracy, but to improve the detection of severe crashes while maintaining reasonable performance for non-severe ones. In this sense, the shift from a decision tree with severe-class recall near zero to a CatBoost multiclass model with severe-class recall around 0.4 is a significant success. We then further improve the prediction of severe and non-severe outcomes by reframing the task as a binary classification problem, allowing the model to focus on the distinction that matters most for real-world safety applications. Under this formulation, CatBoost with oversampling achieves substantially higher severe-class recall (0.68) and an AUC of 0.744, demonstrating strong discriminative ability and confirming that severity aggregation is an effective strategy for high-risk crash identification.

B. Lessons Learned

From this project we learned several important lessons about working with large, imbalanced real-world datasets:

- **Decision trees alone are not enough.** Even with tuning, a single tree tends to overfit majority patterns and fails on rare classes.
- **Gradient boosting is more reliable for complex, non-linear crash patterns.** XGBoost and CatBoost handle the categorical structure and interactions better and offer higher macro performance.
- **Handling imbalance requires multiple strategies.** Oversampling, hybrid methods, and careful metric choices are necessary to avoid being misled by high accuracy.
- **Model interpretability is essential.** Feature importance from tree-based ensembles provides insight into the mechanisms behind severe crashes, highlighting harmful event, collision type, lighting, weather, and intersection relation as key factors.

C. Limitations and Future Work

Our study has several limitations:

- We only use a subset of available variables; richer physical features such as speed, distance to junction, and traffic control timing are not yet incorporated.
- Driver and vehicle details (seatbelt use, vehicle safety ratings, impairment indicators) are not fully modeled but could strongly influence severity.
- Severe crashes remain relatively rare even after oversampling; external years or additional data sources could be used to augment them.

Future work includes:

- Integrating additional variables and external datasets to improve predictive accuracy.
- Exploring deep learning architectures for tabular data and comparing them to boosted trees.
- Deploying the binary severe vs. non-severe model in real-world scenarios such as emergency response prioritization, dynamic risk maps, and roadway safety analysis tools.

D. Overall Summary

Crash severity is inherently difficult to predict due to class imbalance, overlapping conditions, and missing causal variables. However, our models uncover clear real-world patterns. Boosting models significantly improve the detection of severe crashes, raising recall from near-zero in the decision tree baseline to around 40–50% in the multiclass setting and to nearly 70% in the binary setting. Key factors driving severity include harmful event, collision type, lighting, weather, and intersection relation—all consistent with real crash dynamics. The binary CatBoost model (severe vs. non-severe) exhibits strong discriminative ability with an AUC of 0.742, making it a promising component for real-world high-risk crash identification systems.

Acknowledgment

We thank the National Highway Traffic Safety Administration (NHTSA) for providing access to national crash data that made this analysis possible.

References

[1] National Highway Traffic Safety Administration (NHTSA), "Traffic Safety Facts and Crash Data Resources." Available: https://www.nhtsa.gov/

[2] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed. Morgan Kaufmann, 2011.

[3] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011. Available: https://scikit-learn.org/stable/

[4] G. Lemaître, F. Nogueira, and C. K. Aridas, "Imbalanced-learn: A Python Toolbox to Tackle the Curse of Imbalanced Datasets in Machine Learning," *Journal of Machine Learning Research*, vol. 18, no. 17, pp. 1–5, 2017. Available: https://imbalanced-learn.org/stable/